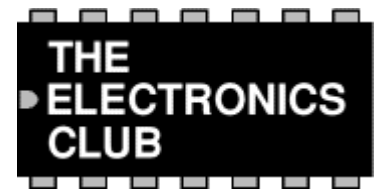


[Home](#) | [Map](#) | [Projects](#) | [Construction](#) | [Soldering](#) | [Study](#) | [Components](#) | [555](#) | [Symbols](#) | [FAQ](#) | [Links](#)

Voltage Dividers

They are also called **Potential Dividers**



Next Page: [Transistor Circuits](#)

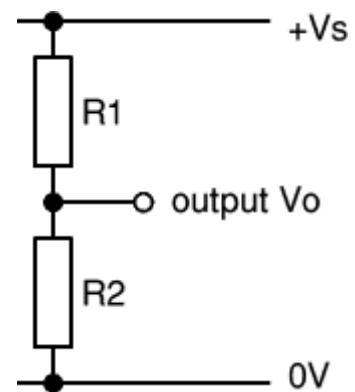
Also See: [Transducers](#) | [Voltage and Current](#) | [Resistance](#) | [Impedance](#)

Voltage divider (potential divider)

A voltage divider consists of two resistances R1 and R2 connected in series across a supply voltage Vs. The supply voltage is divided up between the two resistances to give an output voltage Vo which is the voltage across R2. This depends on the size of R2 relative to R1:

- **If R2 is much smaller** than R1, **Vo is small** (low, almost 0V)
(because most of the voltage is across R1)
- **If R2 is about the same** as R1, **Vo is about half Vs**
(because the voltage is shared about equally between R1 and R2)
- **If R2 is much larger** than R1, **Vo is large** (high, almost Vs)

(because most of the voltage is across R2)



$$V_o = \frac{V_s \times R_2}{R_1 + R_2}$$

If you need a precise value for the output voltage Vo you can use [Ohm's law](#) and a little algebra to work out the formula for Vo shown on the right. The formula and the approximate rules given above assume that negligible current flows from the output. This is true if Vo is connected to a device with a high resistance such as voltmeter or a chip input. For further information please see the page on [impedance](#). If the output is connected to a [transistor](#) Vo cannot become much greater than 0.7V because the transistor's base-emitter junction behaves like a diode.

Voltage dividers are also called **potential dividers**, a name which comes from potential difference (the proper name for voltage).

One of the main uses of voltage dividers is to connect [input transducers](#) into circuits...

Using an input transducer (sensor) in a voltage divider

Most input transducers (sensors) vary their **resistance** and usually a voltage divider is used to convert this to a varying **voltage** which is more useful. The voltage signal can be fed to other parts of the circuit, such as the input to a chip or a transistor switch.

The **sensor** is one of the resistances in the voltage divider. It can be at the top (R1) or at the bottom (R2), the choice is determined by when you want a large value for the output voltage Vo:

- Put the sensor at the **top** (R1) if you want a **large Vo** when the sensor has a **small resistance**.

- Put the sensor at the **bottom** (R2) if you want a **large Vo** when the sensor has a **large resistance**.

Then you need to choose a value for the resistor...

Choosing a resistor value

The value of the resistor R will determine the range of the output voltage Vo. For best results you need a large 'swing' (range) for Vo and this is achieved if the resistor is much larger than the sensor's minimum resistance Rmin, but much smaller than the sensor's maximum resistance Rmax.

You can use a [multimeter](#) to help you find the minimum and maximum values of the sensor's resistance (Rmin and Rmax). There is no need to be precise, approximate values will do.

Then choose resistor value:

$R = \text{square root of } (R_{\min} \times R_{\max})$

Choose a standard value which is close to this calculated value.

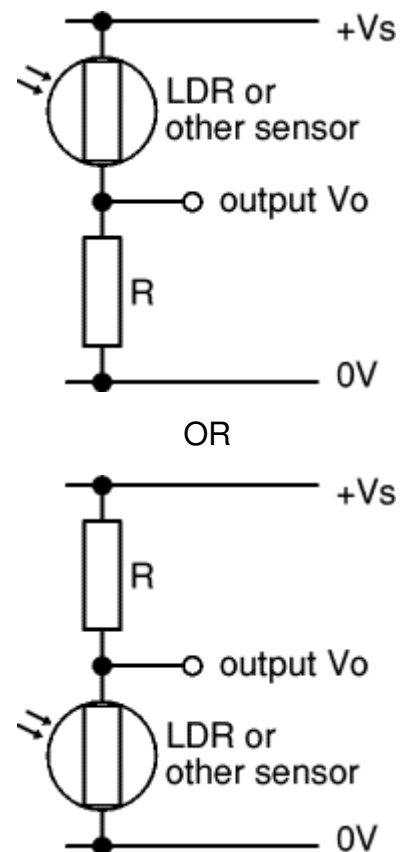
For example:

An LDR: Rmin = 100Ω, Rmax = 1MΩ, so R = square root of (100 × 1M) = 10kΩ.

swapping over the resistor and sensor

The resistor and sensor can be swapped over to invert the action of the voltage divider. For example an LDR has a high resistance when dark and a low resistance when brightly lit, so:

- **If the LDR is at the top (near +Vs),**
Vo will be **low in the dark** and **high in bright light**.
- **If the LDR is at the bottom (near 0V),**
Vo will be **high in the dark** and **low in bright light**.



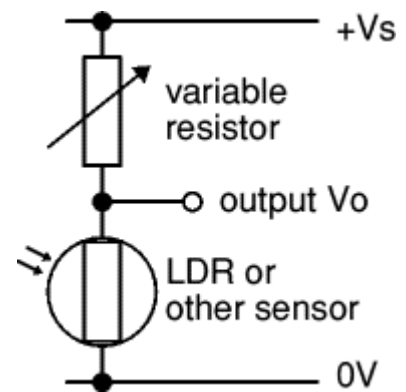
Using a variable resistor

A variable resistor may be used in place of the fixed resistor R. It will enable you to adjust the output voltage Vo for a given resistance of the sensor. For example you can use a variable resistor to set the exact brightness level which makes a chip change state.

The variable resistor value should be larger than the fixed resistor value. For finer control you can use a fixed resistor in series with the variable resistor. For example if a 10kΩ fixed resistor is suitable you could replace it with a fixed 4.7kΩ resistor in series with a 10kΩ variable resistor, allowing you to

adjust the resistance from 4.7k to 14.7k Ω .

If you are planning to use a variable resistor connected between the +Vs supply and the base of a transistor you must include a resistor in series with the variable resistor. This is to prevent excessive base current destroying the transistor when the variable resistor is reduced to zero. For further information please see the page on [Transistor Circuits](#).



The sensor and variable resistor can be swapped over if necessary

Next Page: [Transistor Circuits](#) | [Studying Electronics](#)

- [Electronics Club Home Page](#)
- [Site Map](#)
- [Example Projects](#)
- [Construction of Projects](#)
- [Soldering Guide](#)
- [Study Electronics](#)
- [Electronic Components](#)
- [555 Timer](#)
- [Circuit Symbols](#)
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